CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B Tech)

Semester - II Examination May 2010

B.Tech. (CL/EC/ME) Regular Exam &

B.Tech. (CE/EE/IT) Backlog Examination

CL - 103: Mechanics of Solids

Date: 20.05.2010, Thursday **Time:12.30p.m to03.30p.m** **Maximum Marks:70**

Instructions:

1. All questions are compulsory.
2. Use different answer books for each section.
3. Figure to the right indicates full marks of the question.
4. Use of calculator is allowed.
5. Assume suitable data if required.

Section I

Que. 1.a	Define composition and resolution of forces.	02
Que. 1.b	Explain Unlike Parallel Forces, Collinear Forces and Coplanar Concurrent Forces	03
Que. 1.c	Distinguish between moment and couple.	02
Que. 2.a	Three coplanar concurrent forces act on a body at point O. Determine two additional forces, along OA and OB such that resultant of the five forces is zero. (Fig.1)	04
Que. 2.b	State & prove lami's Theorem.	03
Que. 2.c	Determine the centroid of the plane area shown in Fig.2 with respect to x and y axes as shown in figure.	07
	OR	
Que. 2.a	Find out the resultant of forces shown in Fig.3 with respect to point A.	07
Que. 2.b	Determine the centroid of the plane lamina shown in Fig.4 with respect to point E.	07
Que. 3.a	Determine the forces in all members of the truss shown in Fig.5.	07
Que. 3.b	A 15^0 wedge has to be driven for tightening a body of 1000 N weight as shown in Fig.6. If angle of friction of all surfaces in contact is 18^0, find force P that start movement of block.	07
	OR	
Que. 3.a	Using method of joint, find forces in all the members with their nature for a truss shown in Fig.7.	07

Que. 3.b	A ladder having 500 N weight and 5m length rests on a rough horizontal ground at one end and leans against a rough vertical wall at the other end making an angle of 60^0 with horizontal. The coefficient of static friction at both ends of the ladder is 0.35. A man of weight 700 N climbs up on the ladder. Calculate minimum distance travelled by the man on the ladder when it is on verge of slipping.	07

Section II

Que. 4.a	Explain how you will find hardness of a given metal in the laboratory.	03
Que. 4.b	Write differential equations for undamped and damped single degree of freedom system.	02
Que. 4.c	Define oscillation, amplitude, time period and natural frequency.	02
Que. 5.a	A stepped bar is loaded as shown in Fig.8. Calculate the stresses in each part and total change in the length of the bar. Take Es = 200 GPa, Ec = 100 GPa, Eb = 80 GPa.	03
Que. 5.b	Two brass rods each 10mm diameter and one steel rod of 16mm diameter together support a load of 50 kN as shown in Fig.9. Find stresses in each rods. Take Es = $2*10^5$ N/mm^2, Eb = $1*10^5$ N/mm^2.	04
Que. 5.c	Derive relation between E and K.	07

OR

Que. 5.a	A steel member ABCD, with three different square cross sections as shown in Fig.10 is subjected to concentrated axial loads. Calculate load P for equilibrium and compute the net change in length of the member. E = 200 GPa.	03
Que. 5.b	A load of 2000 kN is applied on a short concrete column 500mm x 500mm reinforced with 4 nos. of 10 mm dia. steel bars. Find stresses in concrete and steel. Take Es = $2.1*10^5$ N/mm^2, Ec = $1.4*10^5$ N/mm^2.	04
Que. 5.c	A steel rod 20 mm in diameter and 200 mm long is heated through 100^0C and at the same time subjected to an axial pull P. If total extension of the rod is 0.30 mm, calculate the magnitude of the pull P. The coefficient of linear expansion of steel is $12*10^{-6}/^0C$ and E = 200 GPa.	07
Que. 6.a	Derive relation between load, shear force and bending moment.	03
Que. 6.b	Draw shear force and bending moment diagrams for a beam shown in Fig.11.	04
Que. 6.c	Draw shear force and bending moment diagrams for a beam shown in Fig.12 indicating values at all important points.	07

OR

Que. 6.a	A beam ABCD in which AB = 2m, BC = 4m and CD = 2m carries a point load of intensity 3 kN each at A and D and UDL of 4 kN/m between B and C. Beam is simply supported at point B and C. Draw shear force and bending moment diagrams.	07
Que. 6.b	Draw shear force and bending moment diagrams for a beam shown in Fig.13 indicating values at all important points.	07

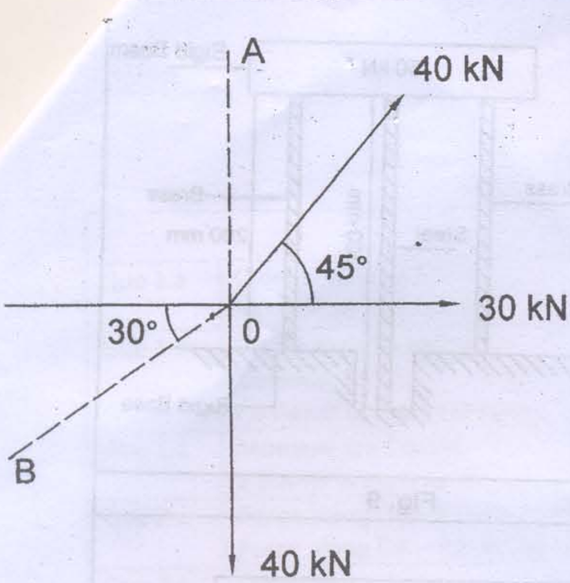

Fig. 1

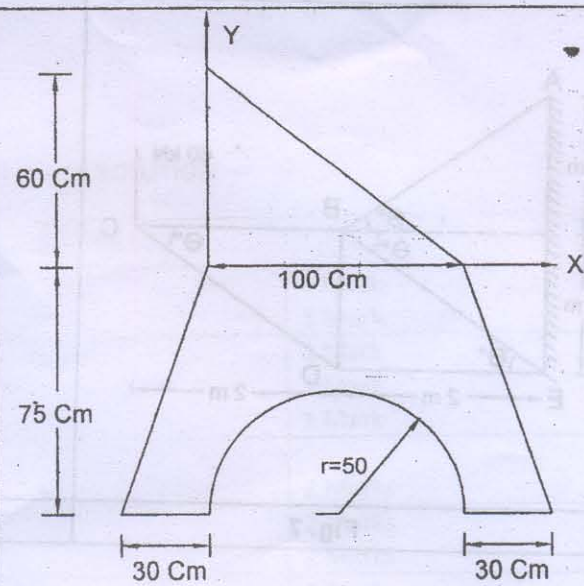

Fig. 2

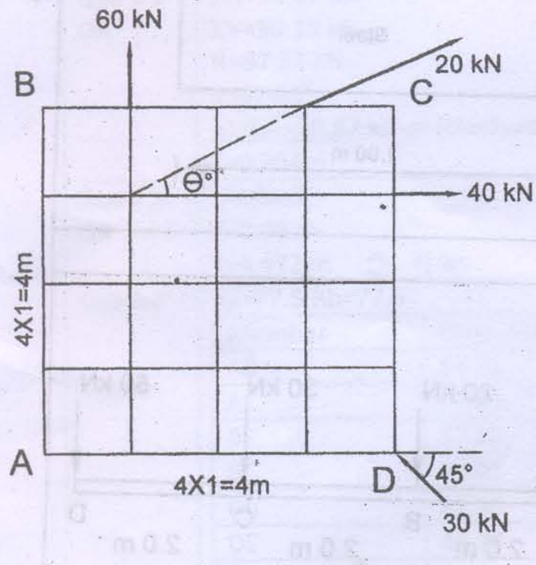

Fig. 3

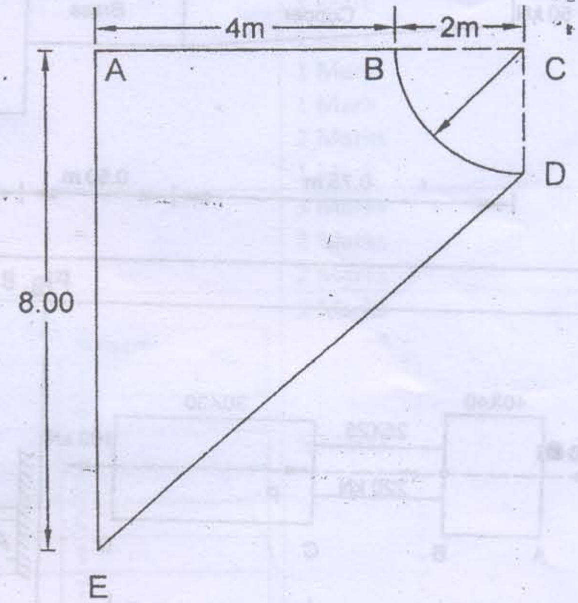

Fig. 4

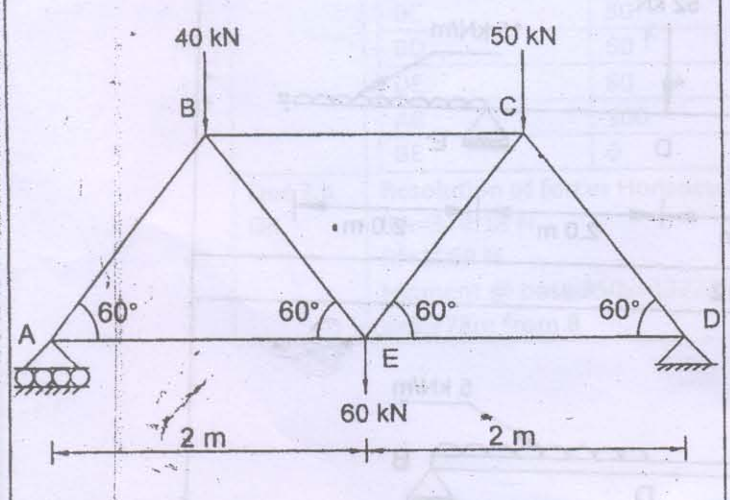

Fig. 5

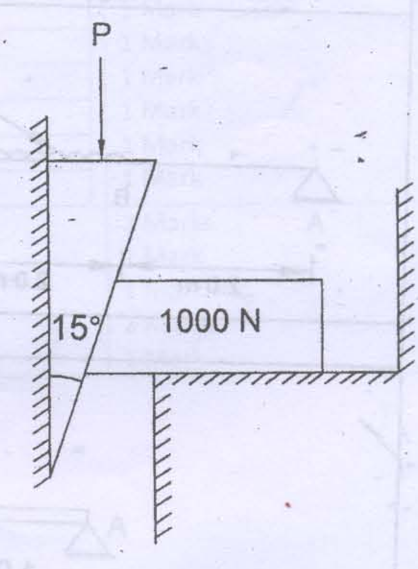

Fig. 6

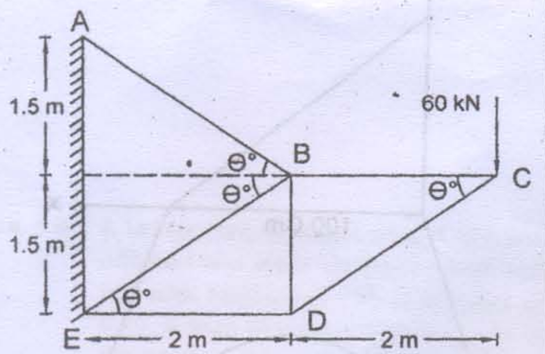

Fig. 7

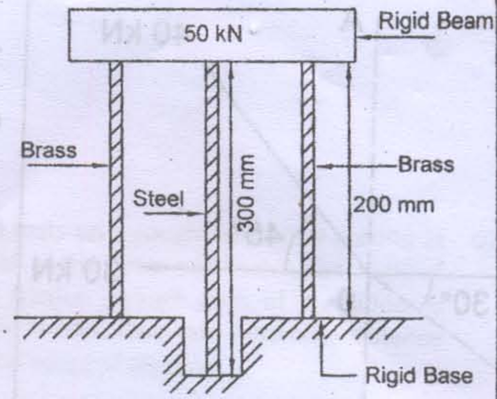

Fig. 9

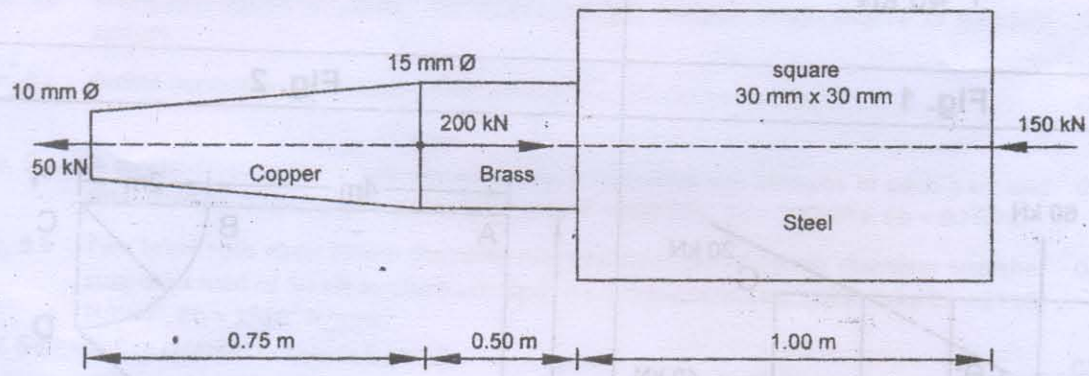

Fig. 8

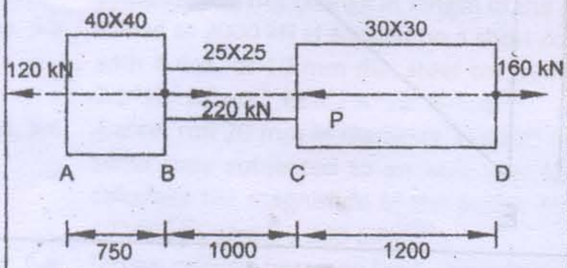

Fig. 10

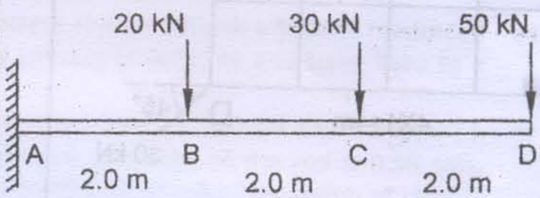

Fig. 11

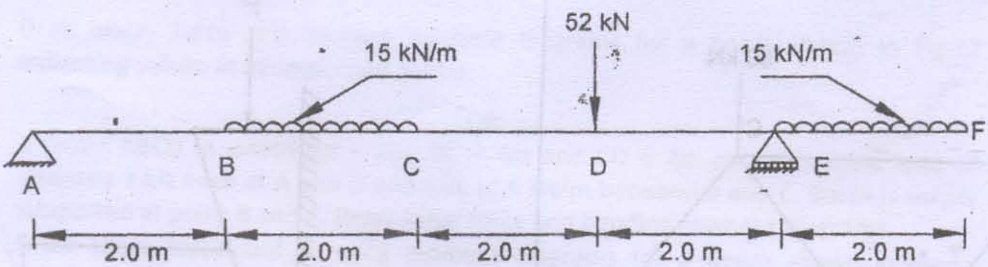

Fig. 12

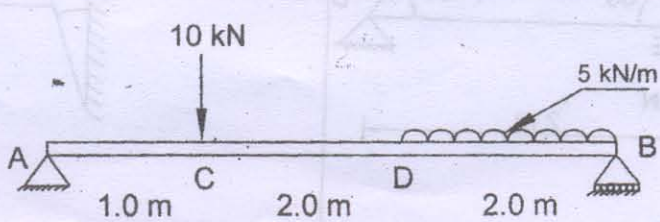

Fig. 13

4 of 4.